

Statistical Data Analysis

Lecture 3: Goodness of fit tests and Kernel density estimation

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Periods 4 & 5

📍 Chapter 3: Exploring distributions

- ◀◀ Section 3.1 – The quantile function and location-scale families
- ◀◀ Section 3.2 – QQ-plots
- ◀◀ Section 3.3 – Symplots
- ◀◀ Section 3.4 – Two-sample QQ-plots
 - Section 3.5 – Goodness-of-fit tests

📍 Chapter 4: Density estimation

- Section 4.1 – Kernel density estimators
- ▶▶ Section 4.2 – Choice of kernel and bandwidth
- ▶▶ Section 4.3 – Cross-validation
- ▶▶ Section 4.4 – Other density estimators

Recap: Hypothesis Testing

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- › We formulate a research question and the corresponding null resp. alternative hypothesis
 - › Notation and terminology: we test H_0 against H_1
 - › Alternative is formulated as the logical negation of the null
 - › Hypotheses are formulated in terms of the population (not the sample)
 - › Null is rejected only if there is enough 'evidence' that contradicts it in the data

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 - › Alternative is formulated as the logical negation of the null
 - › Hypotheses are formulated in terms of the population (not the sample)
 - › Null is rejected only if there is enough 'evidence' that contradicts it in the data
- › From a sample, a test statistic $T = T(X)$ is computed
 - › The test statistic has a certain distribution under H_0
 - › Given a sample x , we look at where $t = T(x)$ falls.
 - › Acceptance region (H_0 is not rejected) vs. critical region
 - › Critical values: boundaries of the acceptance region

Recap: Hypothesis Testing – Significance, p -value

- › Statistically significant result: rejection of H_0
 - › The value of $t(x)$ is very unlikely if H_0 is true.
- ② What is it meant by very unlikely?

Recap: Hypothesis Testing – Significance, p -value

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- Ahead of data collection and testing, the significance level α is set
 - α is the probability of mistakenly rejecting the null hypothesis when it is true (type I error)
 - Typical values of α are 5% or 1%

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- › Ahead of data collection and testing, the significance level α is set
 - › α is the probability of mistakenly rejecting the null hypothesis when it is true (type I error)
 - › Typical values of α are 5% or 1%
- › To reach a test conclusion, we compare α (set a priori) with the p -value
 - › $p\text{-value} = \mathbb{P}(T(X) \text{ is "more extreme" than } t(x) \mid H_0 \text{ is true}).$
 - › $p\text{-value} \geq \alpha \rightsquigarrow$ do not reject H_0
 - › $p\text{-value} < \alpha \rightsquigarrow$ reject H_0
 - › To compute it we need the distribution of $T(X)$ under H_0 .

Recap: Hypothesis Testing – Error types and power

- › The errors made in hypothesis testing are
 - › Type I error: Rejecting H_0 , while it is true
 - › Type II error: Not rejecting H_0 , while H_1 is true.
- › The power of a test is

$$\mathbb{P}(\text{Reject } H_0 \mid H_1 \text{ is true}) = 1 - \mathbb{P}(\text{Type II error})$$

- › We look for tests with $\mathbb{P}(\text{Type I error}) \leq \alpha$ and a large power.

Recap: Hypothesis Testing – Example: Mean Caffeine Content

- › A chemist buys an espresso every morning from the same bakery
 - › Lately, however, she feels more tired than usual
 - › She want to test whether the average caffeine (the **unknown** μ) is significantly less than what the bakery states ($\mu_0 = 65\text{mg}$)
 - › She analyzes caffeine content in her daily espresso for a month ($n = 31$)
 - › For simplicity, we assume **known** standard deviation $\sigma = 4$

Recap: Hypothesis Testing – Example: Mean Caffeine Content

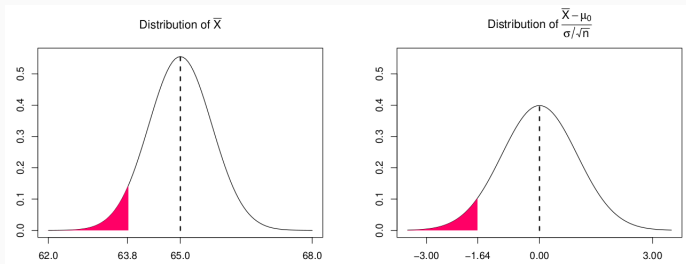
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 - › She analyzes caffeine content in her daily espresso for a month ($n = 31$)
 - › For simplicity, we assume **known** standard deviation $\sigma = 4$
- › **Hypotheses:** $H_0 : \mu \geq 65$ against $H_1 : \mu < 65$
 - › **Test statistic:** the sample mean: $T(X) = \bar{X}$
 - › **Distribution of $T(X)$ under H_0 :**

$$\bar{X} \stackrel{H_0}{\sim} N(\mu_0, \sigma^2/n), \text{ standardized: } \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \stackrel{H_0}{\sim} N(0, 1)$$

- › If σ is estimated via s , $\frac{\bar{X} - \mu_0}{s/\sqrt{n}} \stackrel{H_0}{\sim} t_{n-1} \xrightarrow{n \rightarrow \infty} N(0, 1)$

Recap: Hypothesis Testing – Example: Mean Caffeine Content

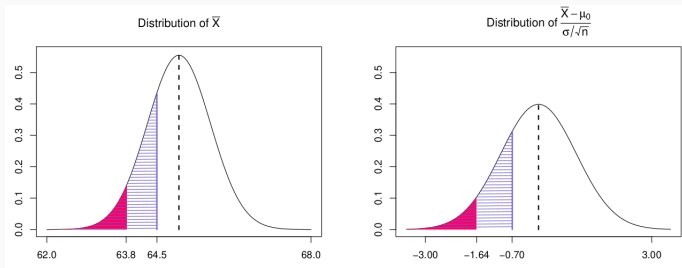
- What is **very unlikely** under the null? **Rejection region** has size $\alpha = 5\%$



- We reject when
 - $\bar{X} \leq 63.8 = \text{qnorm}(\alpha, \text{mean} = \mu_0, \text{sd} = \sigma/\sqrt{n})$
 - $\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \leq -1.64 = \text{qnorm}(\alpha).$

Recap: Hypothesis Testing – Example: Mean Caffeine Content

- › The specific sample collected by the chemist has mean $\bar{x} = 64.5$
- › Compute the test statistic $t = \frac{64.5 - \mu_0}{\sigma/\sqrt{n}} = \frac{64.5 - 65}{4/\sqrt{31}} \approx -0.7$
 - › Falls in the **acceptance region**
 - › **p-value**: probability of observing, under the null, a result *as least as extreme* (here, a sample mean *smaller* than 64)
 - › The **p-value** is $\mathbb{P}(Z < -0.7) = 0.24 > 0.05 = \alpha$, where $Z \sim N(0, 1)$.
- ❓ Do we **reject** or **fail to reject** H_0 ?



Goodness-of-Fit Tests

Motivation

- › How do we find a distribution that is suitable to the data?
 - › Graphical methods
 - › Goodness-of-fit tests
- › Some data clearly resemble well-known distributions, others less so
 - › Let us look at some real data on the [Human Benchmark website](#)
 - › You can also play these cognitive games to see how you compare to others, if you like (but do so after class, please 😊)
 - › **Note:** we do not use these data in our analysis, but  provides us with other data about cognitive tests
- › Using package `languageR`, we will look at reaction times from a **lexical decision** (**word** vs. **non-word**) test. In particular:

```
# Consider only subject T2 and only correctly-answered trials  
subject_T2 <- subset(languageR::lexdec, (Subject == "T2") & (Correct == "correct"))  
# Column RT is the logarithm of reaction times in ms, we transform it  
RT_T2 <- exp(subject_T2$RT)
```

Goodness-of-Fit Tests

- › Idea: sample $x_1, \dots, x_n$ from unknown F and test

$$H_0 : F \in \mathcal{F}_0 \quad \text{vs.} \quad H_1 : F \notin \mathcal{F}_0$$

- › Simple H_0 : $\mathcal{F}_0 = \{F_0\}$ is one distribution
- › Composite H_0 : $\mathcal{F}_0$ is a collection of distributions (e.g., a LSF)

Goodness-of-Fit Tests

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- › Composite H_0 : $\mathcal{F}_0$ is a **collection** of distributions (e.g., a LSF)
- › Three tests: **Develop a test**
 - › Shapiro-Wilk test
 - › Kolmogorov-Smirnov test
 - › χ^2 -test

Shapiro-Wilk Test

› Composite H_0 : $F \in \{N(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma > 0\}$

› Test statistic:

$$W = \frac{(\sum_{i=1}^n a_i X_{(i)})^2}{\sum_{i=1}^n (X_i - \bar{X})^2} \in (0, 1]$$

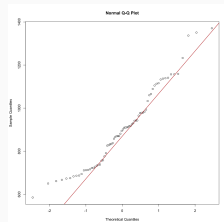
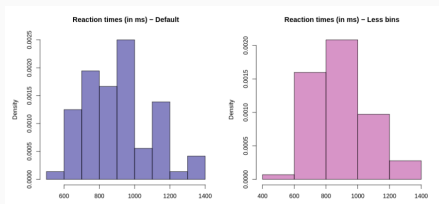
- › a_i known constants associated to normal order statistics and only depend on n , not on the data itself.
- › Under the null, numerator and denominator are both proportional to an estimator for σ^2
- › Otherwise, unrelated $\rightsquigarrow$ reject the null for relatively 'small' values of W

› Distribution of W under H_0

- › Null distribution of W not known in closed form
- › Can be approximated numerically.
- › (Approximate) p -values are given by : `shapiro.test`

Shapiro-Wilk Test – Example

- Do reaction times (correct answers) of R2 look normally distributed?



- A test can give a more **objective** answer

```
shapiro.test(RT_T2)

##
##  Shapiro-Wilk normality test
##
## data:  RT_T2
## W = 0.95398, p-value = 0.01024
```

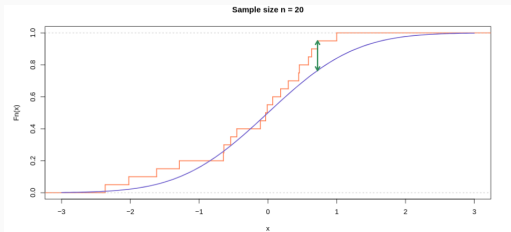
- At significance level $\alpha = 5\%$, do we **reject** the null r not?
- What about $\alpha = 1\%$?

Kolmogorov-Smirnov Test

- › Consider **simple** $H_0 : F = F_0$ (vs. $H_1 : F \neq F_0$)
 - › **Caveat**: not for testing **composite** H_0 (i.e., membership to same LSF)
- › **Test statistic** is the maximum vertical distance between $\hat{F}_n$ and F_0

$$D_n = \sup_x \left| \hat{F}_n(x) - F_0(x) \right|$$

↪ Null rejected for large values of D_n



Kolmogorov-Smirnov Test – Null distribution

> Null distribution of D_n

- > Known in closed form, but not a well-known distribution
- > Depends only on n , not on F_0 (if continuous) $\rightsquigarrow$ nonparametric test
- > p -values are given by : `ks.test`
- > Wrong usage: `ks.test(x, "pnorm", mean(x), sd(x))`

Kolmogorov-Smirnov Test – Example I

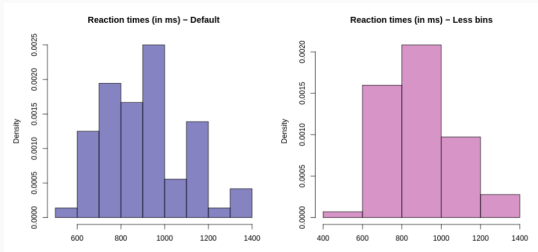
```
set.seed(2102)
x <- rnorm(10^4)

## [1] "Sample mean: -0.006, sample standard deviation: 0.999"
# Correct use of K-S test
ks.test(x, "pnorm", mean = 0, sd = 1)

##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: x
## D = 0.0094869, p-value = 0.3291
## alternative hypothesis: two-sided
# Incorrect use of K-S test
ks.test(x, "pnorm", mean = mean(x), sd = sd(x))

##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: x
## D = 0.0071734, p-value = 0.6822
## alternative hypothesis: two-sided
```

Kolmogorov-Smirnov Test – Example II



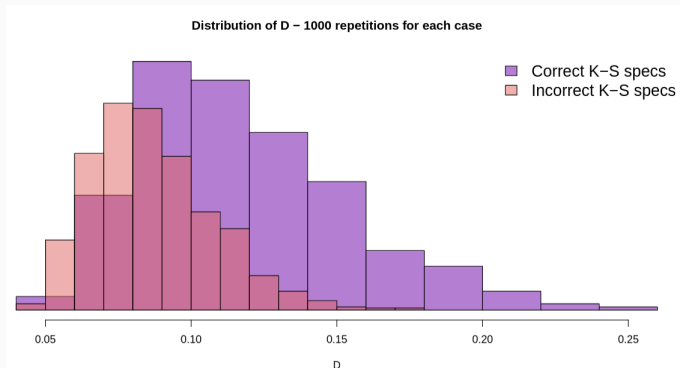
```
ks.test(RT_T2, "pnorm", mean = 900, sd = 120)
```

```
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: RT_T2
## D = 0.18927, p-value = 0.0115
## alternative hypothesis: two-sided
# Compute mean and variance for the sample

## [1] "Sample mean: 903.778, sample standard deviation: 184.023"
# Incorrect use of K-S test
ks.test(RT_T2, "pnorm", mean = mean(RT_T2), sd = sd(RT_T2))
```

```
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: RT_T2
## D = 0.093944, p-value = 0.5488
## alternative hypothesis: two-sided
```

Kolmogorov-Smirnov Test – What Happens?



Recall that we **reject** for **large** values of D . . .

- › Consider **simple** $H_0 : F = F_0$ (vs. $H_1 : F \neq F_0$)
 - › **Caveat**: not for testing **composite** H_0 (i.e., membership to same LSF)
- › Test statistic

$$\chi^2 = \sum_{i=1}^k \frac{[N_i - np_i]^2}{np_i}$$

- › Based on the difference between **observed** and **expected** number of observations in intervals $I_1, \dots, I_k$
- ›

$$N_i = \sum_{j=1}^n 1_{I_i}(X_j) \quad p_i = F_0\{I_i\} = \int_{I_i} dF_0$$



- › Null rejected for large values of χ^2
- › Null distribution of χ^2 is asymptotically χ_{k-1}^2 (reliable if all $np_i \geq 5$)
 $\rightsquigarrow$ **distribution-free** (asymptotically)

Comparison of the tests

› Shapiro-Wilk test

- › Only for (composite) test $H_0 : F \in \{N(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma > 0\}$.
- › Test statistic compares estimators for σ^2
- › `shapiro.test`

› Kolmogorov-Smirnov test

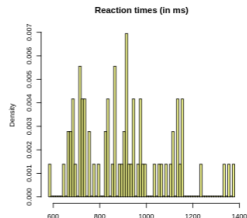
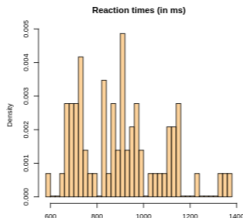
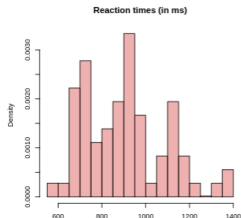
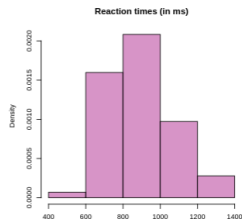
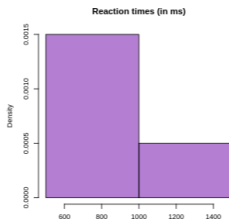
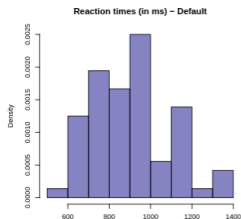
- › Only for simple hypothesis $H_0 : F = F_0$.
- › Test statistic computes largest distance between F_n and F_0 .
- › Distribution-free
- › `ks.test`

› χ^2 -test

- › Only for simple hypothesis $H_0 : F = F_0$
- › Test statistic compares observed and expected observations within bins
- › Asymptotically distribution-free
- › `chisq.test`

Kernel Density Estimation

Histograms as Density Estimators?



Reaction times (in ms) for correct answers given by subject T2 (from `languageR::lexdec`)

Arbitrary bin widths and/or break choices $\leadsto$ visualisation can be misleading

Kernel Density Estimation – Idea

- › X has density f if

$$P(a < X < b) = \int_a^b f(t)dt \quad \text{for all}$$

- › **Density estimation**: reconstruction of true density based on data
- › Sample $x_1, \dots, x_n$: realizations of i.i.d. random variables $X_1, \dots, X_n$ from unknown (continuous) distribution F with density f
- › **Aim**: nonparametric density estimator $t \rightarrow \hat{f}(t)$ (stochastic function)

Kernel Density Estimation – The histogram

- › The histogram as density estimate, with bin width h

$$\hat{f}(t) = \frac{1}{n} \sum_{i=1}^n \frac{1}{2h} 1_{[x_i-h, x_i+h]}(t)$$

- › For each t , count the number of data points within the bin width.
- › Alternative formulation:

$$\hat{f}(t) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h} K\left(\frac{t - x_i}{h}\right), \quad \text{with } K(x) = \frac{1}{2} 1_{[-1,1]}(x)$$

- › Note: K is the density of $\text{Unif}(-1, 1)$, with mean zero .

Kernel Density Estimation – Definition

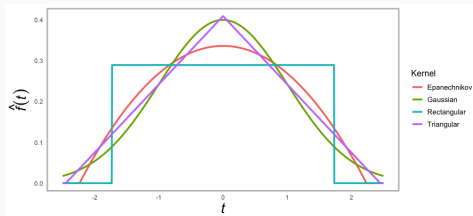
- › Kernel density estimator:

$$\hat{f}(t) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h} K\left(\frac{t - x_i}{h}\right)$$

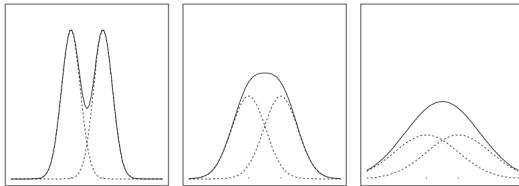
- › $\hat{f}$ distributes mass $1/n$ smoothly around each x_i , according to **kernel** K and with spread of mass specified by **bandwidth** $h > 0$
- › **Kernel** K is a given density with expectation 0 and variance 1

Choice of Kernel and Bandwidth

- › How to choose K and h ?
- › Kernel choice is less critical:



- › Bandwidth choice is more critical:



Gaussian kernel, two observations, different bandwidths $\rightsquigarrow$ very different estimators

Kernel Density Estimation – in R

- › `plot(density(x))` and `ggplot(aes(x=x)) + geom_density()`
 - › Automatically select a bandwidth based on data and Gaussian kernel.
 - › Add keywords `kernel = "rectantular"` and `bw = 12`
- › Access values by `density(x)$y`

Summary and wrap up

- › **Goodness-of-fit tests** give us hypothesis tests to test whether data originates from a given (family of) distribution(s).
 - › Shapiro-Wilk test
 - › Kolmogorov-Smirnov test
 - › χ^2 -test
 - › All have different strengths and weaknesses.
- › **Kernel density estimation** is used to reconstruct the true density based on data (nonparametrically)
 - › Kernel
 - › bandwidth
 - › Choice of bandwidth is more critical
 - › More on density estimation next week!

- › What now?
 - › Continue working on assignment 2
 - › Deadline: 24 February 23:59
 - › All students should be working in pairs now.
- › Final announcement
 - › Tutorial groups 1 and 4 (11:00) are merged into 1 group at 11:00 in NU-3A67